Piyush Bhalwalkar

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PROFESSIONAL SUMMARY

B.Tech. graduate in Information Technology (9.3 CGPA) specializing in full-stack MERN development. Published two npm packages (piyushai, piyush17). Experienced in building real-time systems, WebRTC video calling, ML-powered sign-language captioning, and payment gateway integration.

EDUCATION

B.Tech. in Information Technology — Mumbai University (SAKEC)	**CGPA: 9.3 / 10**	**2021 – 2025**
HSC — Maharashtra Board (S.I.W.S.)	**92.50%**	**2021**
SSC — Maharashtra Board (J.W.E.H.S.)	**77.20%**	**2019**

TECHNICAL SKILLS

Languages:	Java, HTML, CSS, JavaScript
Frameworks:	React, Node.js, Express.js, Socket.io
Databases:	MySQL, Redis
Tools & Protocols:	Git, Postman, REST APIs, Google Translate API, Web Speech Synthesis
Core Concepts:	Data Structures & Algorithms, DBMS, SDLC

PROJECTS

SHUZ — E-Commerce Platform | shuz.onrender.com

Technologies: MongoDB, Express.js, Node.js, React (MERN Stack)

- Built a full-stack e-commerce platform for sneaker heads with a React frontend and RESTful Node.js/Express backend.
- Implemented JWT-based user authentication, Redis caching for all products listing, persistent shopping cart, and third-party payment gateway.
- Built an admin dashboard with role-based access for product listing, inventory updates, and order management.

Kisaan Saathi — AI Farming Assistant | ksaathi.onrender.com

Technologies: React, Python, Flask, MongoDB, Gemini AI, Weather Map API

- Built a full-stack AI-powered farming assistant for Indian farmers with a React frontend and Flask REST API backend.
- Integrated Google Gemini AI to generate crop-specific fertilizer recommendations, farming plans, and pest/disease advice.
- Implemented multi-language support (English, Hindi, Marathi) with real-time weather data.
- Designed a farm session flow collecting soil data (pH, soil type) to produce personalized AI-driven crop recommendations.

RTSLD — Real-Time Sign Language Detection | rtsld.onrender.com

Technologies: Node.js, Express.js, Socket.io, Python, KNN, Web Speech Synthesis, WebRTC

- Built a platform that converts sign language gestures into live captions during video calls using a KNN model trained on hand-landmark data.
- Integrated Web Speech Synthesis and Google Translate API for multilingual audio output (Hindi, Marathi, English) with minimal latency.
- Implemented peer-to-peer video calling and screen sharing via WebRTC with Socket.io handling signaling and room management.
- Designed an UI that overlays real-time captions on the video stream for an accessible experience for hearing-impaired users.

CERTIFICATIONS

- JavaScript Essentials 1 (JSE) — Cisco Networking Academy
- MERN Stack Internship — SAKEC